



What is the Medical Student Experience of Incivilities on the Surgery Clerkship and Do They Matter?

Gazi Rashid, MD,^{*} Esther Kim, MD,[†] Amy Evenson, MD,[‡] Reza Askari, MD, PhD,[§] Emil Petrusa, PhD,^{||} David Rogers, MD, MHPE,[¶] and Roy Phitayakorn, MD, MHPE^{||}

^{*}Department of Surgery, Inova Fairfax Hospital, Fairfax, Virginia; [†]Department of Surgery, University of Miami Hospital and Clinics, Ft Lauderdale, Florida; [‡]Department of Surgery, Beth Israel Deaconess Medical Center, Boston, Massachusetts; [§]Department of Surgery, Brigham and Women's Hospital, Boston, Massachusetts; ^{||}Department of Surgery, Massachusetts General Hospital, Boston, Massachusetts; and [¶]Dept. of Surgery, University of Alabama in Birmingham, Birmingham, Alabama

OBJECTIVE: To study medical students' experiences with incivilities on the surgery clerkship

DESIGN: A retrospective cross-sectional survey study based on the Workplace Incivility Scale 12 (WIS-12) that quantified the frequency of experienced incivilities, averaged the frequency of all twelve incivilities as an index, and characterized how they affected students' perceptions of surgery

SETTING: A single large urban medical school affiliated with multiple quaternary level medical centers

PARTICIPANTS: From October 2022 to 2023, all medical students on their surgery clerkship at 3 different medical centers received a paper survey near the end of the 12-week clerkship period. The response rate was 83% (118/144), of whom 41% self-identified as male and 38% as White.

RESULTS: Ninety-seven percent of respondents experienced an incivility during the clerkship, the most frequent of which was receiving little attention for their opinions (3.07 ± 1.34 , mean $\pm$ standard deviation), being interrupted (2.36 ± 1.38), and being ignored (2.27 ± 1.37). Students whose interest in a surgical career decreased by the end of the clerkship had a significantly higher index of incivility than those whose interest increased or remained stable (2.13 ± 0.86 vs. 1.70 ± 0.58 respectively, $p < 0.005$). Students whose perception of surgical culture worsened by the end of the clerkship had a significantly higher index compared to those whose perceptions improved or remained stable (2.12 ± 0.71 vs. 1.68 ± 0.59 respectively, $p < 0.05$).

Correspondence: Inquiries to Gazi Rashid, MD, Department of General Surgery, Inova Fairfax Hospital, 7701 Woodmont Ave, Apt 1106, Bethesda, MD 20814; e-mail: gazirashid@gmail.com

CONCLUSIONS: Medical students reported incivilities as nearly ubiquitous during the surgery clerkship. A higher frequency of these experiences was associated with decreased interest in surgery and worse perceptions of surgical culture. Surgical educators should track and intervene on these behaviors during the clerkship. Other specialties should study these phenomena as well. (J Surg Ed 82:103535. © 2025 Association of Program Directors in Surgery. Published by Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.)

KEY WORDS: clerkship, incivilities, medical students, mistreatment, surgery, WIS-12

COMPETENCIES: Interpersonal and Communication Skills

INTRODUCTION

Incivilities are undesirable workplace behaviors that pervade healthcare. In the workplace psychology literature, they are defined as low-intensity behaviors that violate the norms of respect and have ambiguous intent to harm. These qualities of intent and intensity distinguish incivilities from more egregious mistreatment, such as abuse, harassment, or bullying.¹

Incivilities clearly affect various healthcare professions, but only recently have been studied systematically in academic surgery.² A 2022 survey of 16 institutions by Dr. Santosa and colleagues found that >90% of surgery faculty and residents have experienced at least 1 workplace incivility in the past year. They also found that a

higher burden of incivilities was experienced by women and Asian Americans.³ Furthermore, incivilities have been shown to harm the workplace at both an environmental and an individual level. Experiencing or even witnessing these behaviors can negatively affect workplace culture, team communication, and even clinical performance.^{1,3-7} For students, experiencing incivilities can hamper learning.^{8,9} In a qualitative analysis of interviews with 42 medical students, experiencing or even witnessing uncivil behaviors in the operating room (OR) created a stressful environment and impaired performance and learning.¹⁰ Incivilities—especially those related to gender—are also associated with developing burnout and have been shown to impact decisions during specialty selection.^{2,11} Incivilities have similar effects at higher positions of power and can be associated with withdrawal from work or “quiet quitting” in surgical attendings and residents.³

Despite our growing understanding about their prevalence and harm, incivilities toward medical students have not been systematically studied in surgery or as experiences distinct from other types of mistreatment. A 2023 scoping review of “incivilities” in medical student education found that most of the studies focused on behaviors that can be considered more egregious, such as abuse or bullying. Those studies that did investigate incivilities as classically defined did so in nonsurgical specialties or examining singular actions such as being neglected.¹¹ Surgery clerkships may have higher reported rates of mistreatment compared to other clerkships. The extent to which incivilities affect medical students may represent a blind spot in our understanding of the surgical learning environment.¹²

In this study, we investigated how medical students experience incivilities during their surgery clerkship. Specifically, we aimed to describe the frequency and characteristics of these interactions and explore how the students perceived them and how they affected their perceptions of surgery. We hypothesized that these behaviors are highly prevalent and associated with worse perceptions of and interest in surgery.

METHODS

Overview

We performed a retrospective cross-sectional survey at a single urban medical school with 3 primary clinical sites, each of which is a large academic medical center with separate Departments of Surgery. This research was deemed exempt by our Institutional Review Board.

Setting

Medical students were surveyed during their surgery clerkship from October 2022 to October 2023. A 1-time paper survey was administered during the last 2 to 3 weeks of their 12-week surgery clerkship. To minimize in-person bias, it was administered in-person by research fellows who were not affiliated with the medical school or the hospitals; they also exited the classroom after providing the surveys and returned only to collect the forms.

Survey Development

A 31-item survey was developed and is included in [Appendix A](#). The survey asked demographic questions, the length of prior clerkship experience, and completion of the obstetrics/gynecology (Ob/Gyn) rotation. These questions were asked to determine if prior clinical experience, especially time spent in the OR environment and with OR teams during Ob-Gyn, was associated with differences in rates of and perceptions of incivilities.

The survey had items adapted from the Workplace Incivility Scale 12 (WIS-12, [Fig. 1](#)), a well-known measure that lists twelve incivilities and asks on a Likert scale how frequently each behavior was experienced over a year.¹³ We modified the timescale to our 3-month clerkship period ([Fig. 2](#)). We also characterized the setting and sources of these experiences, allowing each option to be scored separately as present or absent. Respondents could select all that apply. We then evaluated how perceptions of surgery and interest in a surgical career changed over the course of the clerkship. Students could provide free text responses to why they thought these behaviors occurred to them and how they could be addressed. The survey was piloted and modified through cognitive interviewing of 6 medical students who had completed the surgery clerkship. These responses were not included in the study analysis.

Index of Incivilities

For each respondent, an index was calculated by averaging the reported frequency (on [Fig. 2](#) Likert scale) of all twelve behaviors from the WIS-12. The range was from 1 to 5. This index was adopted from Santosa and colleagues, who developed and calculated this metric and utilized it in multiple studies exploring incivilities among surgical attendings and residents.^{3,14} It is meant to represent an overall burden of incivilities experienced by an individual.

Statistical Analyses

To maximize sample size, all surveys were included in the analysis, even if not all questions were answered.

Figure 1: The 12 incivilities listed on WIS-12
Paid little attention to your statements or showed little interest in your opinions
Doubted your judgement on a matter over which you had responsibility
Gave you hostile looks, stares, or sneers
Addressed you in unprofessional terms, either publicly or privately
Interrupted or "spoke over" you
Rated you lower than you deserved on an evaluation
Yelled , shouted, or swore at you
Made insulting or disrespectful remarks about you
Ignored you or failed to speak to you (e.g., gave you the "silent treatment")
Accused you of incompetence
Targeted you with anger outbursts or "temper tantrums "
Made jokes at your expense

FIGURE 1. The 12 incivilities listed on WIS-12.

Demographics, experiences on the surgery clerkship, and frequency of experiences were analyzed with descriptive statistics. Bivariate analysis was used to determine if the index of incivilities was significantly different based on distinct factors using a 2-sample t-test and 1-way ANOVA when appropriate. A $p < 0.05$ was considered statistically significant for all tests. All statistical analyses were performed on Stata 17. Free text responses underwent content analysis by 2 reviewers independently, who then deliberated and finalized the results.

RESULTS

From 2022 to 2023, 144 students across 4 clerkship cohorts were surveyed in person. The survey was partially or completely submitted by 118 students (83% response rate). Forty one percent of the respondents identified as male and 38% identified as White (Table 1).

Frequency of Incivilities

Of the respondents, 97% experienced at least 1 incivility during the clerkship. Out of the 87 respondents who had completed clerkships prior to surgery, 68% said incivilities occurred "more often" or "far more often" on

surgery than other clerkships, while 24% said it was "equivalent" and 8% said they occurred "less often" or "far less often" on surgery. If students experienced an incivility, they were asked to indicate which groups were sources and where they occurred. The 3 groups

TABLE 1. Breakdown of the Characteristics and Index of Incivilities of Respondents

Characteristics	n	% of Respondents	Index of Incivilities mean (SD), * $p < 0.05$
Gender			
Female	55	49	1.75 (0.66)
Male	46	41	1.7 (0.61)
NonBinary	2	2	1.68 (0.96)
Prefer not to answer	10	9	1.98 (0.74)
Total	113	100	
Race			
African American	4	4	1.83 (0.75)
Asian American	34	30	1.71 (0.6)
Hispanic, Latin-X, Spanish	9	8	1.75 (0.7)
White	42	38	1.69 (0.3)
Prefer not to answer	13	12	2.02 (0.65)
Other	10	9	1.82 (0.77)
Total	112	100	
Completed Ob-Gyn already?			
No	60	51	1.82 (0.61)
Yes	58	49	1.78 (0.69)
Total	118	100	
Prior clerkship exp			
None	29	25	1.82 (0.61)
3-6 months	53	45	1.77 (0.71)
7-9 months	36	31	1.79 (0.66)
Total	118	100	

* indicates $p < 0.05$.

Figure 2: Frequency Likert scale in our modified WIS-12
1: Never
2: Once or Twice Overall
3: Once or Twice per Month
4: Once per Week
5: Many Times per Week

FIGURE 2. Frequency Likert scale in our modified WIS-12.

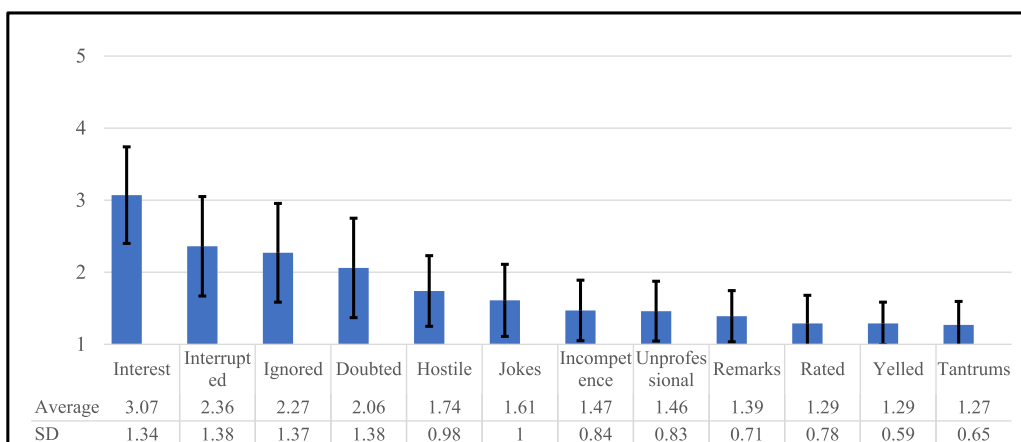


FIGURE 3. Frequency of incivilities across all students (average score on Likert scale and standard deviation).

most frequently listed as sources of these behaviors were residents (indicated by 85% of respondents), attendings (75%), and scrub techs/circulators (62%). Perioperative areas (i.e., OR) were the most frequently listed setting (87%), followed by the workroom (44%), and the wards (39%).

Figure 3 demonstrates the frequency of all twelve behaviors averaged across all respondents, along with the standard deviation. The most frequently experienced behaviors included receiving little interest or attention for their opinions (3.07 ± 1.34 , mean $\pm$ standard deviation), being interrupted or spoken over (2.36 ± 1.38), being ignored or given the silent treatment (2.27 ± 1.37), and being doubted on a matter on which they had responsibility (2.06 ± 1.38). The other 8 behaviors all had an average frequency of less than 2, which represented “Once or Twice Overall.” The 4 most frequent incivilities had the highest standard deviations, while the standard deviation of the other 8 incivilities was less than 1.

For all respondents, the index of incivilities was 1.79 ± 0.67 , which falls between “never” and “once or twice overall.” As seen on Table 1, bivariate analysis of the index based on gender or race demonstrated no significant differences between any groups. There was also no significant difference in the index of incivilities based on length of prior clerkship experience or completion of the Ob/Gyn clerkship.

Possible Effects of Incivilities

To examine the possible consequences of incivilities, students were asked how strongly they felt these behaviors influenced their impressions of surgery (Table 2). Forty five percent of respondents ($n = 51$) said that experiencing incivilities had a “moderately” or

“extremely” strong influence in their interest in a surgical career, while 58% ($n = 65$) said that incivilities had a “moderately” or “extremely” strong influence on their perceptions of surgical culture.

Bivariate analyses demonstrated that an overall negative change in students’ impressions of surgery was associated with a higher index of incivilities (Table 3). Students whose interest in a surgical career “decreased” over the course of the clerkship had a significantly higher index of incivility than those whose interest “increased or remained stable” (2.13 ± 0.86 vs. 1.70 ± 0.58 , $p < 0.005$). Students whose perception of surgical culture worsened by the end of the clerkship also had a significantly higher incivility index versus those whose perceptions improved or did not change (2.12 ± 0.71 vs. 1.68 ± 0.59 , $p < 0.05$).

TABLE 2. Influence of Experiencing Incivilities on Perceptions of Surgery

How Strongly Did Your Experiences With These Behaviors...	n	% of Total
...Influence your interest in surgery?		
Not at all	27	24
Mildly	35	31
Moderately	32	28
Extremely	19	17
Total	113	100
...Influence your perception of culture in surgery?		
Not at all	12	11
Mildly	35	31
Moderately	34	30
Extremely	31	28
Total	112	100

TABLE 3. Association of Index of Incivilities with Changes in Perceptions of Surgery

Over the Course of the Surgery Clerkship	n	% of Total	Index of Incivilities Mean (SD) *p < 0.05
Your perception of surgical culture...			
Remained the same or improved	91	86	1.68 (0.59)
Got worse	15	14	2.12 (0.71)*
Total	106	100	
Your interest in surgical career...			
Remained the same or increased	93	79	1.69 (0.58)
Decreased	25	21	2.13 (0.86)*
Total	118	100	

* indicates p<0.05.

Possible Sources of and Solutions Towards Incivilities

Students were also asked why they thought these experiences occurred (Table 4). Content analysis of the free text responses found 4 major categories of responses. The most common category (48.2% of responses) was “Displacement of Stressors,” where stressors such as being exhausted or burned out, working in a high-stakes environment, and difficulties outside of work were underlying reasons for incivilities. The “Culture of Surgery,” such as a culture that tolerated these behaviors, or a hierarchical environment with stark power dynamics, were the second most common category of incivilities (23.1%). “Individual Characteristics of the Source” such as personality traits and “Attitudes Toward Medical Students” such as not being considered part of the team were the third and fourth most common categories (15.8% and 10.8%, respectively).

When students were asked about how these behaviors can be addressed (Table 5), content analysis found 3 major categories, 2 of which focused on prevention through systemic (36.8%) and team-level changes (29.9%). On a system level, students proposed changing recruiting practices to improve diversity, along with reducing stress in the workplace. Team-level changes included clarifying expectations, such as making the surgical team aware of what was expected of medical students and addressing incivilities from OR scrub techs and nurses. The last category was responding to incivilities that have occurred (26.4%), including instituting meaningful consequences for these actions, and not assigning students to known offenders.

DISCUSSION

Incivilities not only pervade all levels and professions of surgery, but they also degrade workplace performance and culture. Our study was the first to systematically quantify and characterize incivilities toward medical students on the surgery clerkship. These experiences are ubiquitous for our medical students (a 97% prevalence), which was similar to the surgical literature which found that 92% of responding surgical residents and attendings experienced at least 1 incivility over the measured time period.³ This prevalence has also been demonstrated also in the broader healthcare literature.^{1,11}

When examining the individual behaviors experienced, we found that only 4 of the twelve behaviors assessed in our study occurred more than once or twice overall on average. These 4 behaviors also had the highest standard deviations (all greater than 1.3), indicating a wide variability among students’ experiences. Interestingly, the burden of incivilities is independent of student gender, ethnicity, length of prior clerkship experience or prior experience with Ob/Gyn. The findings in this study are different from prior studies in several ways. Prior work, including Santosa and colleagues, found that significantly more incivilities were reported by surgical attendings and residents who identified as female or Asian-American.^{3,14}

The lack of associations with individual characteristics in our study could indicate that the experience of incivilities for medical students is dependent more on external factors such as the residents or attendings on a service, the nurses that work in a particular OR, or the situation surrounding the incivility. While it is possible that our lack of race- or gender-based difference may be due to a disparity in sample size, we can corroborate these findings in our other results. Our content analysis of the reasons that students thought they experienced incivilities mostly focused on external sources such as overworked and exhausted residents and faculty, a hierarchical culture, etc. Less than 5% of responses focused on individual traits such as race or gender or other internal causes. In contrast, prior work found that 33.8% of surgical attendings and residents chose gender as a potential cause, while 49.1% of them collectively selected other traits, like age, appearance, race, religion, etc.¹² These differences from our study suggest that the underlying perceived drivers of incivilities and thus, potential solutions to address them, may be individualized to the recipient or differ by role they inhabit. As Generation Z learners (i.e., born mid 1990s-early 2010s) enter medical school and residency, future research should also examine generational attitudes towards incivilities to better understand these differences between how students, residents, and faculty.

TABLE 4. Content Analysis of Responses to “If you Experienced the Behaviors Listed Above on Your Surgery PCE, What do you Think is the Reason for These Experiences?”

Categories (Number of Responses, %)	Sub-Categories	Number of Responses	% of Responses	Example Responses
Displacement of stressors (n = 67, 48.2%)	Overworked or burned out	41	29.5	“Because people are taking out their own exhaustion, arrogance, or state of being unwell on others”
	High-stakes environment	14	10	“It’s a stressful field with stressful moments so people can be on edge sometimes”
	General	10	7.2	“Displaced stress/frustration”
	outside of work	2	1.4	“Home life not good”
	Permissive culture	11	7.9	“They are tolerated, and people will behave inappropriately if they are allowed to do so”
	Perpetuating the cycle	9	6.4	“I also think when people are trained in a toxic culture, they perpetuate it”
	Hierarchy and Power Dynamics	7	5.1	“Even felt like residents were different 1-on-1 vs. w/ residents or attendings, like they had to act a certain way”
Culture of surgery (n = 33, 23.1%)	General	5	3.6	“Younger residents tend to be kinder and get harsher as [they] get older”
	Personality traits	8	5.8	“Surgery has tended to attract a personality type focused on achievement above all and they judge others the same”
	Poor communication and teaching styles	6	4.3	“Scrub nurse was trying to protect sterility but doesn’t know how to deliver that feedback in a kinder, more helpful way”
	Emotional immaturity	4	2.9	“Have low self-reflection/control over behavior”
Individual characteristics of the source (n = 22, 15.8%)	General	4	2.9	“Undiagnosed/untreated mental illness”
	Not part of the team, So not worth the time	5	3.6	“Also, students are transient, so it’s easy to ignore/dismiss them since we have limited time on each team”
	Lowest on the totem pole	4	2.9	“I think there is a power dynamic unique to scrub techs and medical students where scrub techs feel medical students are the only people in the room, they have power over, which can lead to mistreatment”
	Incompetent	3	2.2	“Expectations that I am incompetent”
	General	3	2.2	“perception that the surgery rotation is supposed to be ‘tough’ on med students”
Miscellaneous (n = 3, 2.2%)	General	3	2.2	“social privilege (this is absolutely associated with race, sex, sexuality; never occurred with nonmale and nonwhite individuals)”

The ultimate impact of incivility for some medical students may be a worse perception of the surgical department and a subsequent decreased interest in pursuing a surgical specialty. Notably, the index of incivility of both groups whose perceptions of and interest in surgery became worse were 2.12 and 2.13, which are relatively low in frequency. These results could indicate that even a “low” index—i.e., a few experiences over 12 weeks—may be enough to have an effect. Even if these experiences do not affect perception of or interest in surgery, incivilities may

affect the educational value of the clerkship. Commonly described situations in our content analysis included poor integration into the team due to a variety of reasons that ranged from a simple lack of introductions to the perception that the team viewed students as “incompetent” and/or “nonessential” due to their transient roles.

These types of situations and triggers for incivilities are also described in the literature as behaviors of both active and passive neglect.^{9,10,15,16} A systematic review of neglect by Kloos and colleagues defined

TABLE 5. Content Analysis for Responses to “How do you Think Experiences With These Behaviors can be Addressed or Improved on the Surgery PCE?”

Categories (Number of Responses, %)	Sub-Categories	Number of Responses	% of Responses	Illustrative Example Responses
Preventing incivilities through systemic changes (n = 32, 36.8%)	Reducing workplace stress	16	18.4	“Don’t overwork the residents to the point of exhaustion. Find ways to reduce the stressful OR work environment (i.e., control whatever can be controlled).”
	Changing the work environment	6	6.9	“Overall culture change”
	Changing hiring practices	5	5.7	“Improved diversity within surgery - all acts of mistreatment... occurred from individually who didn’t look like me.”
	Raising awareness of mistreatment	2	2.3	“Honest communication about how prevalent such mistreatment can be”
Preventing incivilities through Team-level changes (n = 26, 29.9%)	Miscellaneous	3	3	“Change American healthcare”
	Taking inclusive actions	7	8	“Basic intros and acknowledgement of everyone would be helpful”
	Addressing students’ relationship with or staff	6	6.9	“Professionalism treatment training for scrub techs/nurses; Explanation that they are in a teaching hospital. . .”
	Understanding, communicating medical students’ expectations	6	6.9	“Make hospital staff/residents aware of the nonclerkship obligations for medical students”
	Training in education and leadership	5	5.7	“... train residents on how to work with and teach med students, even those uninterested in surgery”
	Miscellaneous	2	2.3	“Assign a team member to be med student point person”
				“Appropriate response to student reports of behavior/treatment”
Reporting and responding to incivilities (n = 23, 26.4%)	Demonstrating accountability through meaningful consequences	11	12.6	
	Improving reporting mechanisms	5	5.7	“Anonymous reporting, but this is hard because sometimes there’s only 1 med student or junior for the entire block”
	Identifying and diverting students from known sources	5	5.7	“Do not assign medical students to preceptors known to be prone to outbursts and mistreatment; asking the residents and coordinator, it is clear that there are a couple of attendings most responsible for most of this behavior and their identity is widely known”
	Responding in real-time	2	2.3	“People should feel comfortable calling out the behavior in the moment or bringing said concerns to surface.”
Miscellaneous (n = 6, 6.9%)	General	6	6.9	“Not sure”

neglect as “the mistreatment of medical students by the medical care team via a lack of meaningful inclusion in the clinical environment such that it has a notable negative impact on learning and student well-being, regardless of intentionality.”¹⁷ The review found various reasons why neglect may occur, including microaggressions, prioritizing efficiency, the

brevity of clerkships limiting familiarity, and being labeled as disengaged if they had to attend to other clerkship responsibilities. In addition, the largest perceived reasons for incivilities in our study was related to stressors experienced by the sources of incivilities. Moving forward, system-level changes to the medical education ecosystem (shorter preclinical time,

decreased surgery clerkship length, increased clinical training sites, etc.) should be carefully analyzed for increased possible medical student and trainee neglect and their experiences with incivilities. Future research could also focus on the well-being of sources like residents and attendings and how their stress and burnout relate to downstream incivilities towards medical students.

Medical students offered many potential solutions to incivilities on the surgery clerkship. These are broadly divided into 2 categories: prevention and reaction. In terms of prevention, there were many feasible and simple ideas, such as making students feel more included, making surgery teams aware of the students' other clerkship responsibilities, and raising awareness of incivilities. An intervention in line with the latter is the Civility, Respect, and Engagement in the Workforce (CREW) intervention, which was developed by the Veteran's Administration.¹⁸ CREW raises overall awareness of the harms of incivilities and help colleagues develop a shared mental model of civility. Positive outcomes of CREW included improving work relationships and decreasing burnout.¹⁹

In response to incivilities that have occurred, students felt that reporting mechanisms were insufficiently anonymous and/or lacked consequential changes. This belief may also be reflected in the "prefer not to answer" groups in gender/race demographics who had a higher corresponding index of incivilities value. Although these differences did not reach statistical significance, it is possible that a larger powered study would reflect this feeling of lack of anonymity when reporting incivilities. Current literature supports this sentiment, demonstrating that there are low rates of reporting among medical students despite awareness of reporting mechanisms because "incidents didn't seem important enough to report" and "fear of reprisal."¹¹ Furthermore, current reporting policies and mechanisms do not specifically include incivilities. The survey developed in this study could be used to surveil these behaviors. Based on those results, specific rotations and sites can be addressed and low-scoring attendings could be offered faculty development. Another proposed solution was identifying residents and faculty that are known sources of incivilities and not assigning them learners, sparing students a poor experience on the clerkship and improving its educational value.

There were several limitations to this study. Although this study is from a single medical school, our participants completed rotations at surgical departments of 3 different hospitals which could improve the generalizability of our results. Our findings demonstrate that behaviors of incivility that are distinct from gross mistreatment do occur to medical

students across different hospitals. Another limitation is that this is a retrospective cross-sectional study that relies on self-reporting. As we only surveyed students, we did not include the perspective of potential sources of incivilities, such as attendings or residents. Lastly, it is important to note that the definitions of mistreatment and incivility are variable and overlap within the literature. However, no previous literature found by these authors has studied a collective group of incivilities amongst medical students in this context.

CONCLUSION

In our study, nearly all medical students on their surgical clerkship experienced incivilities. Furthermore, which groups experience a higher burden of incivilities may be different for students compared to surgical residents and attendings. Ultimately, these experiences degrade the learning environment and are associated with decreased interest in surgery and worse perceptions of surgical culture. Surgical educators should take steps to understand and address these behaviors, including developing interventions to prevent them from occurring or building monitoring and reporting programs to address them when they do occur.

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SUPPLEMENTARY INFORMATION

Supplementary material associated with this article can be found in the online version at [doi:10.1016/j.jsurg.2025.103535](https://doi.org/10.1016/j.jsurg.2025.103535).